Test Series No.

02

NEET Success Kit [2026-27]

DURATION: 195 Minutes | **M. MARKS: 720**

DATE: 10/06/2026

SYLLABUS

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Botany:	Full Syllabus
Zoology:	Full Syllabus

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 15 minutes** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

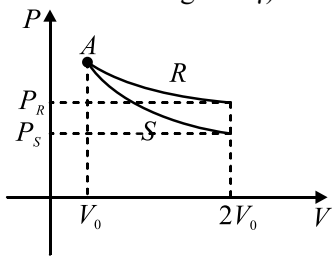
Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. An ideal gas is initially at thermodynamic state A as shown in PV diagram. It is separately expanded by two processes in which process R is isothermal and process S is adiabatic such that the final volume in both the cases is same and it is double of initial volume (V_0). The ratio of final pressure $\frac{P_R}{P_S}$ is (adiabatic index of the gas is γ):



(1) γ (2) $2^{\gamma-1}$

(3) 2^{γ} (4) $2^{1-\gamma}$

2. A bulb with rated parameters of 250 W, 220 V is connected to a 110 V dc supply. The electrical energy consumed in one second by the bulb is:

(1) 62.5 J (2) 125 J

(3) 225 J (4) 100 J

3. An ideal liquid exhibits steady flow through a horizontal tube of varying cross-section. The widest and narrowest parts of the tube have diameters of 7.5 cm and 3.0 cm, respectively. Find the ratio of the minimum fluid velocity to the maximum fluid velocity in the tube.

(1) $\frac{25}{4}$ (2) $\frac{4}{25}$

(3) $\frac{2}{5}$ (4) $\frac{5}{2}$

4. The variation of intensity (I), of light with distance (r) from the axis of an infinite cylindrical source is given by:

(1) $I \propto \frac{1}{r^2}$ (2) $I \propto \frac{1}{r}$

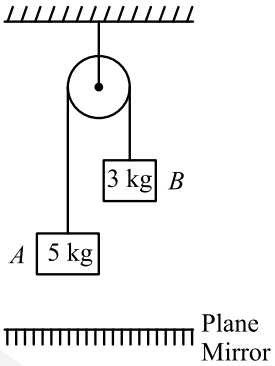
(3) $I \propto r$ (4) $I \propto \frac{1}{\sqrt{r}}$

5. The capacitance of a parallel plate capacitor charged to a potential V and having vacuum in between the plates is C. What is the electrical energy possessed by an imaginary cube of side a present inside the capacitor? (Area of each plate is A)

(1) $\frac{CVa^3}{2\varepsilon_0 A}$ (2) $\frac{CV^2a^3}{2\varepsilon_0{}^2A^2}$

(3) $\frac{C^2V^2a^2}{2\varepsilon_0 A^2}$ (4) $\frac{C^2V^2a^3}{2\varepsilon_0 A^2}$

6. A system of two blocks A and B connected to an ideal fixed pulley is released from rest at time $t = 0$. What is the speed of image of block A in the plane mirror fixed horizontally on the ground at time $t = 2$ s? ($g = 10$ m/s^2) (Assume block does not hit the mirror before $t = 2$s)



(1) 10.0 m/s

(2) 5.0 m/s

(3) 2.5 m/s

(4) 1.25 m/s

7. A ray of light enters a rectangular glass slab of refractive index $\sqrt{3}$ at an angle of incidence 60°. It travels a distance of 4 cm inside the slab and emerges out of the slab. The perpendicular distance between the incident and the emergent rays is:

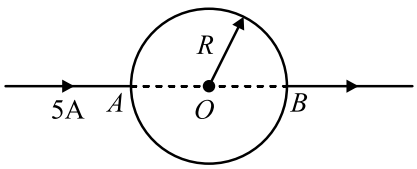
(1) 4 cm

(2) 2 cm

(3) $4\sqrt{3}$ cm

(4) $2\sqrt{2}$ cm

8. The current flowing in a straight long conductor is 5A. At point A the current divides into two semicircular loops of radius 1m. The electric resistance of one of the semicircle is 3Ω and that of the other is 2Ω. What is the net magnetic field at the common centre O?



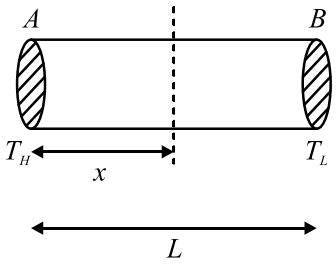
(1) Zero

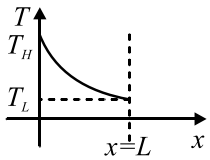
(2) $\pi \times 10^{-7}$ T

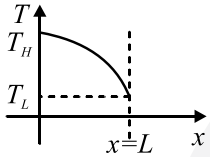
(3) $2\pi \times 10^{-7}$ T

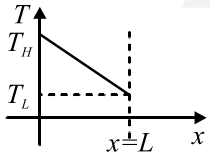
(4) $\frac{\pi}{2} \times 10^{-7}$ T

9. Heat is flowing steadily in a uniform cylindrical rod with the ends maintained at temperature T_H and T_L ($T_H > T_L$) as shown below. The temperature (T) of the cross-section at a distance x from end A varies as:



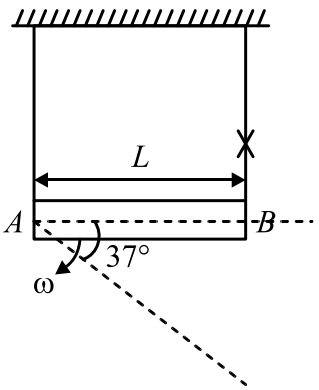
(1) 

(2) 

(3) 

(4) Data insufficient

10. A uniform horizontal rod AB of mass m and length L is suspended in horizontal position with the help of two strings as shown in the figure. The string supporting the end B is cut. What is angular speed (ω) of the rod when it makes an angle of 37° with the horizontal? (Assume the string attached at end A will remain vertical during the motion of rod, acceleration due to gravity is g)



(1) $\omega = \sqrt{\frac{3g}{2L}}$ (2) $\omega = \sqrt{\frac{4g}{L}}$

(3) $\omega = \sqrt{\frac{3g}{8L}}$ (4) $\omega = \sqrt{\frac{9g}{5L}}$

11. The difference 0.205 – 0.2014 can be expressed as (with regards to significant figure)

(1) 0.0036

(2) 3.6×10^{-3}

(3) 4.0×10^{-3}

(4) 0.004

12. At resonance in series RLC a.c. circuit the:

(1) impedance of the circuit is maximum

(2) current in the circuit is minimum

(3) voltage and current in the circuit have zero phase difference.

(4) power factor of the circuit is zero.

13. At what distance from the mean position of a particle executing SHM along x-axis with amplitude A the speed of the particle is half of the maximum speed?

(1) $\frac{A}{2}$ (2) $\frac{A}{4}$

(3) $\frac{\sqrt{3}A}{2}$ (4) $\frac{3A}{4}$

14. The binding energies of the nuclei A and B are E_a and E_b respectively. Three nuclei of the element B fuse to give single nucleus of element A and an energy Q is released. Then E_a, E_b and Q are related as:

(1) $E_a - 3E_b = Q$

(2) $3E_b - E_a = Q$

(3) $E_a + 3E_b = Q$

(4) $E_b + 3E_a = Q$

15. Two runners start simultaneously from the same point on a circular track of radius $50\ \text{cm}$. They run in the same direction with speeds of $25\pi\ \text{cm/s}$ and $20\pi\ \text{cm/s}$. After how many seconds will they meet for the first time?

(1) 40 s (2) 10 s

(3) 50 s (4) 20 s

16. Fifth overtone of a closed organ pipe is found to be in unison with the third overtone of an open organ pipe. Find ratio of lengths of open organ pipe to that of closed organ pipe.

(1) $\frac{5}{11}$ (2) $\frac{7}{11}$

(3) $\frac{4}{11}$ (4) $\frac{8}{11}$

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17. A body cools from 80° to 50°C in 6 minutes. The time it takes to cool from 60°C to 40°C is: (the temperature of surrounding is 20°C)

(1) 6 minutes (2) 10 minutes
(3) 9 minutes (4) 4 minutes

18. Which of the following is **not** an assumption of kinetic theory of gases?

(1) Space occupied by molecules is much smaller than volume of gas.
(2) There is force of interaction between the molecules.
(3) Pressure of the gas is due to elastic collision of gas molecules with the walls of the container.
(4) The molecules are moving randomly in all directions.

19. Consider the following logic gate configurations and evaluate their outputs:

I. An AND gate with inputs 0 and 1.
II. A NOR gate with inputs 0 and 1.
III. A NAND gate with inputs 1 and 1.
IV. An OR gate with inputs 0 and 1.

Which of the following statements is **correct** regarding the outputs of these gates?

(1) Gates I and III have an output of 1, while II and IV have an output of 0.
(2) Gate IV has an output of 1, while gates I, II, and III have an output of 0.
(3) Gates II and IV have an output of 1, while I and III have an output of 0.
(4) All gates have an output of 1.

20. Acceleration due to gravity on the surface of earth is *g*. In **List-I** different values of acceleration due to gravity are given and in **List-II** different values of height and depth from earth's surface are given. Match the **List-I** with **List-II**. (*R* is radius of earth)

List-I		List-II	
(A)	$\frac{g}{2}$	(I)	Depth = R
(B)	$\frac{g}{4}$	(II)	Depth = $\frac{R}{2}$
(C)	$\frac{g}{9}$	(III)	Height = $2R$
(D)	Zero	(IV)	Height = R

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-IV, D-I
(2) A-II, B-IV, C-III, D-I
(3) A-I, B-II, C-IV, D-III
(4) A-IV, B-III, C-I, D-II

21. A conductor having cuboidal shape, has dimensions in ratio 1 : 2 : 5, the ratio of maximum to minimum resistance across two opposite faces is:

(1) 5 : 1 (2) 25 : 1
(3) 10 : 1 (4) 5 : 2

22. The excess pressure inside one soap bubble is 3 times than inside another soap bubble, then the ratio of their respective surface areas is:

(1) 1 : 9 (2) 1 : 27
(3) 1 : 3 (4) 6 : 1

23. The work done by a force $\vec{F} = (-6x^3\hat{i})$ N in displacing a particle from $x = 4$ m to $x = -2$ m is:

(1) – 240 J (2) 360 J
(3) 420 J (4) – 360 J

24. 100 g ice at 0°C is mixed with 100 g water at 80°C. The final temperature of the mixture is:

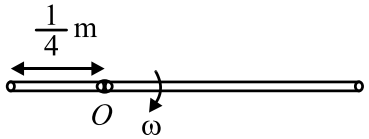
(1) 40°C (2) 25°C
(3) 0°C (4) 47.14°C

25. Choose the option containing **correct** statements:

A. Bulk modulus is relevant for solids, liquids and gases.
B. Shear modulus is only relevant for liquids.
C. Young's modulus is relevant for solids only.

(1) A Only (2) A and B only
(3) C only (4) A and C only

26. A uniform rod of length 1 m and mass 2 kg is hinged at point *O* as shown. If it rotates with angular speed 12 rad/s, then kinetic energy of the rod is:



(1) 42 J (2) 9 J
(3) 81 J (4) 21 J

27. The magnetic flux across a loop of resistance 10 Ω is given by $\phi = (5t^2 - 4t + 1)$ Wb. The charge flown in the loop from $t = 0$ to $t = 2$ s is:

(1) 1 C
(2) 1.2 C
(3) 0.6 C
(4) 2.4 C

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28. A resistor and an inductor are connected in series to a (50 Hz, 200 V), a.c. supply. The current in the circuit is 4 A. If the power consumed in the circuit is 80 watt, the inductive reactance of the circuit is:

(1) $\frac{3}{2}\sqrt{15}\,\Omega$

(2) $\frac{3}{2}\sqrt{25}\,\Omega$

(3) $\frac{5}{2}\sqrt{25}\,\Omega$

(4) $15\sqrt{11}\,\Omega$

29. The time period of revolution of electron of a hydrogen atom in its first excited state is 12.8×10^{-16} s. The frequency of revolution of the electron in its second excited state is:

(1) $2.3 \times 10^{16}\ s^{-1}$

(2) $2.3 \times 10^{12}\ s^{-1}$

(3) $2.3 \times 10^{14}\ s^{-1}$

(4) $2.3 \times 10^{10}\ s^{-1}$

30. A boy of mass 45 kg is standing stationary with respect to horizontal conveyor belt that is accelerating at 2 m/s^2. The magnitude of net force acting on the boy is:

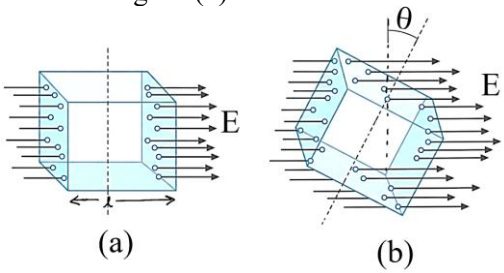
(1) Zero

(2) 90 N

(3) 45 N

(4) 60 N

31. A cube of edge length l is placed in a region of uniform horizontal electric field of magnitude E. Initially, as shown in Figure (a), one pair of opposite faces of the cube is perpendicular to the electric field lines. The cube is then rotated by an angle θ about a axis perpendicular to the electric field lines, as shown in Figure (b).



What is the total magnitude of the electric flux entering the cube in this new rotated position?

(1) $E \cdot l^2\sqrt{1-\sin(2\theta)}$ (2) $E \cdot l^2\sqrt{1+\cos(2\theta)}$

(3) $E \cdot l^2\sqrt{1+\sin(2\theta)}$ (4) $E \cdot l^2\sqrt{1-\cos(2\theta)}$

32. A constant electric field $\vec{E} = E_0(\hat{i}+\hat{j})$ N/C is present in space. There are two points A and B in the x-y plane with coordinates (2, 2)m and (3, 4)m respectively. Choose the **correct** option.

(1) electric potential of A > electric potential of B

(2) electric potential of A < electric potential of B

(3) electric potential of A = electric potential of B

(4) Data insufficient

33. Two point charges A and B, having charges $+Q$ and $-Q$ respectively, are placed at a certain distance apart and force acting between them is F. If 20% charge of A is transferred to B, then force between the charges become:

(1) $\frac{9F}{16}$

(2) $\frac{4F}{5}$

(3) $\frac{16F}{25}$

(4) $\frac{3F}{4}$

34. In a glass capillary, water rises to a height of 6 cm (angle of contact 0°). In the same tube, an unknown liquid L with a density 9 times that of water, falls by 2 cm (angle of contact 120°). The ratio of surface tension of water to surface tension of liquid L is:

(1) 1:3

(2) 3:1

(3) 1:2

(4) 1:6

35. The percentage change in length of 1 m iron rod if its temperature changes by 100°C is (α for iron is 2×10^{-5}/°C)

(1) 0.2%

(2) 2%

(3) 20%

(4) 0.02%

36. A simple microscope consists of a convex lens of power +5 D. If the image is formed at infinity, then the magnifying power is:

(1) 1.25

(2) 2.5

(3) 2.25

(4) 5

37. A beam of light of wavelength 500 nm from a distant source falls on a single slit 1 mm wide and the resulting diffraction pattern is observed on a screen 2 m away. The distance between the first dark fringes on either side of central bright fringe is:

(1) 20 mm (2) $\frac{1}{2}$ mm

(3) 2 mm (4) 1 mm

38. If the dimensions of a physical quantity are given by $M^aL^bT^c$, then the physical quantity will be:

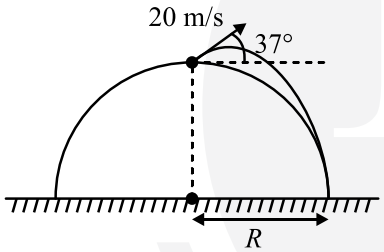
(1) Pressure, if $a = 1, b = -1, c = -2$

(2) Velocity, if $a = 1, b = 0, c = -1$

(3) Acceleration, if $a = 1, b = 1, c = -2$

(4) Force, if $a = 0, b = -1, c = -2$

39. A ball is thrown obliquely at an angle of 37° with the horizontal from top of a hemisphere such that at landing it strikes at a point on the circumference of the circular base of the hemisphere. What is the radius of the hemisphere? ($g = 10$ m/s^2)



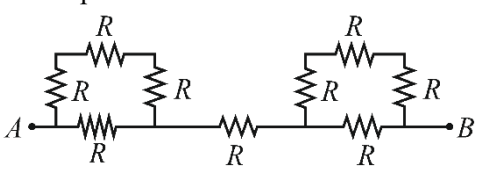
(1) $\frac{448}{10}$ m

(2) $\frac{224}{10}$ m

(3) $\frac{448}{5}$ m

(4) $\frac{64}{5}$ m

40. Nine resistances each of resistance R are connected in a circuit as shown. The equivalent resistance between points A and B is:



(1) $\frac{5R}{2}$ (2) $3R$

(3) $\frac{3R}{2}$ (4) $4R$

41. The linear momentum of a photon having energy 9 MeV is:

(1) 0.01 eV s m^{-1}

(2) 0.02 eV s m^{-1}

(3) 0.03 eV s m^{-1}

(4) 0.04 eV s m^{-1}

42. Two plano-convex lens of focal lengths 20 cm and 40 cm are placed together to form a double convex lens. The resultant focal length will be:

(1) 40 cm

(2) 20 cm

(3) $\frac{40}{3}$ cm

(4) $\frac{20}{3}$ cm

43. A capacitor has a capacity of 5 μF. Electric potential across the capacitor changes with time at the rate of 10^7 V s^{-1}. The displacement current is:

(1) 1 A

(2) 0.5 A

(3) 5 A

(4) 50 A

44. A car starts from rest and moves with constant acceleration on a straight road. The ratio of displacements of the car in first 2 seconds and next 2 seconds after that is:

(1) 1 : 2 (2) 1 : 1

(3) 1 : 3 (4) 4 : 1

45. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The work done by kinetic frictional force is always negative on any particle in a system.

Reason R: Kinetic frictional force always acts opposite to the direction of motion of a particle in a system.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

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46. Total number of stereoisomers for the compound $CH_3-CH=CH-\underset{\underset{CH_3}{|}}{CH}-CH=CH_2$ is:

(1) 8 (2) 4
(3) 6 (4) 2

47. Given below are two statements:

Statement I: Fullerenes are the only pure form of carbon.

Statement II: Tin and lead are hard metals with high melting points.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

48. Consider the following statements:

(A) Enantiomers have different melting point.
(B) Acetone is a polar protic solvent.
(C) CH_3MgBr is an organometallic compound.

Identify the **incorrect** statements:

(1) A and B only (2) A and C only
(3) B and C only (4) A, B and C

49. Consider the reaction;

$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$,

The moles of NH_3 formed by the reaction between 4 moles of N_2 and 9 moles of H_2 is:

(1) 4 mol (2) 7 mol
(3) 6 mol (4) 5 mol

50. For which of the following ions, 'spin only' magnetic moment is maximum?

(1) Cr^{3+} (2) Fe^{3+}
(3) Sc^{3+} (4) Ti^{3+}

51. In acidic medium, consider the reaction between $KMnO_4$ and $FeSO_4$.

Choose the **correct** set of statements:

A. In acidic medium, MnO_4^- is reduced to Mn^{2+}.
B. Each mole of $KMnO_4$ accepts 5 moles of electrons in acidic medium.
C. One mole of $KMnO_4$ oxidizes 5 moles of Fe^{2+} to Fe^{3+} in acidic medium.
D. The oxidation number of Mn decreases from +7 to +4 in acidic medium.

(1) A, B and C are correct
(2) A and C are correct
(3) B and D are correct
(4) A, B, C and D are correct

52. The wavenumber for the shortest wavelength transition in Balmer series of hydrogen atom spectrum is: (Given: R is Rydberg constant)

(1) $\frac{5R}{36}$ (2) $\frac{36R}{5}$
(3) $\frac{4}{R}$ (4) $\frac{R}{4}$

53. **Incorrect** statement among the following is:

(1) when a system is in equilibrium, the entropy is maximum.
(2) the energy of an isolated system is constant.
(3) a spontaneous process is an irreversible process.
(4) Born-Haber cycle is used to measure enthalpy of combustion.

54. Match **List-I** with **List-II**.

List-I (Reaction)		List-II (Major product)	
(A)	$C_2H_5Cl \xrightarrow{KCN}$	(I)	C_2H_5ONO
(B)	$C_2H_5Cl \xrightarrow{AgCN}$	(II)	$C_2H_5NO_2$
(C)	$C_2H_5Cl \xrightarrow{AgNO_2}$	(III)	C_2H_5CN
(D)	$C_2H_5Cl \xrightarrow{KNO_2}$	(IV)	C_2H_5NC

Choose the **correct** answer from the options given below:

(1) A-III; B-IV; C-I; D-II
(2) A-III; B-IV; C-II; D-I
(3) A-IV; B-III; C-II; D-I
(4) A-IV; B-III; C-I; D-II

55. Which of the following is a redox reaction?

(1) $BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$
(2) $Cl_2 + 2KI \rightarrow 2KCl + I_2$
(3) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$
(4) $SiO_2 + 4HF \rightarrow SiF_4 + 2H_2O$

56. A first-order reaction has half-life $t_{1/2} = 5$ min. Select the **correct** statements regarding this:

A. $k = 0.1386\ min^{-1}$.
B. Time for the concentration to become $1/8^{th}$ of initial is 15 min.
C. Time for 75% completion is 10 min.
D. Doubling initial concentration doubles $t_{1/2}$.

(1) A, B and C are correct
(2) A and B only are correct
(3) B and D only are correct
(4) A, B, C and D are correct

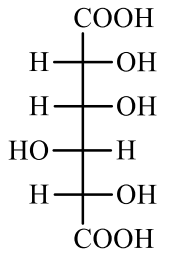
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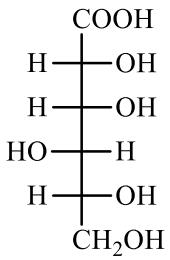
57. The IUPAC name of $[CrCl_3(NH_3)_3]$ is:
(1) Trichloridotriamminechromium(III)
(2) Triamminetrichloridochromate(III)
(3) Triamminetrichloridochromium(III)
(4) Trichloridotriamminechromate(III)

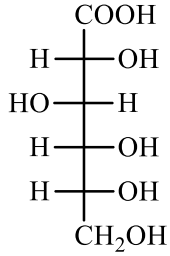
58. Given below are two statements:
Statement I: Phenol gives bubbles of CO_2 on reaction with aq. $NaHCO_3$.
Statement II: Picric acid is less acidic than phenol.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

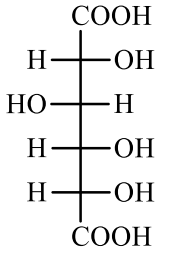
59. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: O_2 molecule is paramagnetic in nature.
Reason R: O_2 molecule contains two unpaired electrons in $\pi 2p_x$ and $\pi 2p_y$ molecular orbitals.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

60. The **correct** representation of saccharic acid is:

(1)


(2)


(3)


(4)


61. During the titration of $H_2C_2O_4$ (oxalic acid) vs $KMnO_4$, which substance acts as an indicator?
(1) $H_2C_2O_4$ (2) $KMnO_4$
(3) Methyl orange (4) EDTA

62. Identify the major product C formed in the following reaction sequence:

$$CH_3CH_2OH \xrightarrow{PCl_5} A \xrightarrow{KCN} B \xrightarrow{H_3O^+} C$$

(1) CH_3COOH (2) $CH_3CH_2CH_2OH$
(3) CH_3CH_2COOH (4) CH_3CH_2CHO

63. If K_1 and K_2 are equilibrium constants for:
$A(g) + B(g) \rightleftharpoons C(g) \ldots K_1$
$C(g) + B(g) \rightleftharpoons D(g) \ldots K_2$
The equilibrium constant for the reaction
$A(g) + 2B(g) \rightleftharpoons D(g)$ is:
(1) $K_1 + K_2$ (2) K_1K_2
(3) $\frac{K_1}{K_2}$ (4) $\frac{K_2}{K_1}$

64. Molality of an aqueous solution in which mole fraction of solute is 0.2, is:
(1) 11.25 m (2) 13.88 m
(3) 12.12 m (4) 14.96 m

65. Energy of which orbital is maximum among the given orbitals for hydrogen atom?
(1) 3d (2) 4s
(3) 3p (4) 2s

66. Arrange the following elements in increasing order of negative electron gain enthalpy:
P, S, Cl, F
Choose the **correct** answer from the options given below:
(1) P < S < F < Cl (2) P < F < S < Cl
(3) S < P < Cl < F (4) F < P < Cl < S

67. Consider the following statements:
A. In BF_3, boron is sp^2 hybridized and the molecule is planar.
B. In NH_4^+, nitrogen is sp^3 hybridized and there is no lone pair on N.
C. In CO, bond order is 3 and the molecule is diamagnetic.
D. In O_2, bond order is 3 and the molecule is diamagnetic.
Choose the **correct** option:
(1) A, B, C are correct
(2) A, B, D are correct
(3) A and B only are correct
(4) A, B, C, D are correct

[Contd...

68. The entropy change in surroundings when 1 mol of $H_2O(l)$ is formed under standard conditions is:
[Given: $\Delta_f H^o$ of $H_2O(l)$ is –286 kJ mol^{-1}]
(1) 959.73 JK^{-1} (2) 848.48 JK^{-1}
(3) 124.62 JK^{-1} (4) 402.64 JK^{-1}

69. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: van't Hoff factor of ethanoic acid in benzene is less than one.
Reason R: Molecules of ethanoic acid dimerise in benzene due to hydrogen bonding.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

70. Which of the following species does **not** show disproportionation reaction?
(1) NO_2^- (2) SO_3^{2-}
(3) ClO^- (4) CrO_4^{2-}

71. Match **List-I** with **List-II**.

List-I (Name of compound)		List-II (Chemical formula)	
(A)	Chile saltpetre	(I)	Na_3AlF_6
(B)	Epsom salt	(II)	PbS
(C)	Galena	(III)	$MgSO_4 \cdot 7H_2O$
(D)	Cryolite	(IV)	$NaNO_3$

Choose the **correct** answer from the options given below:
(1) A-IV; B-III; C-II; D-I
(2) A-II; B-I; C-IV; D-III
(3) A-III; B-I; C-II; D-IV
(4) A-I; B-III; C-IV; D-II

72. For the quantitative estimation of nitrogen in organic compounds, Kjeldahl method is applicable for:
(1) Pyridine (2) Nitrobenzene
(3) Aniline (4) Azobenzene

73. The number of σ bonds in the product obtained on passing ethyne through red hot iron tube at 873 K is:
(1) 6 (2) 9
(3) 15 (4) 12

74. Solubility of $Ni(OH)_2$ in a 0.1 M $Ca(OH)_2$ is:
[Given: K_{sp} of $Ni(OH)_2$ is 2×10^{-15}]
(1) 1×10^{-14} M (2) 5×10^{-14} M
(3) 5×10^{-7} M (4) 1×10^{-7} M

75. Three solutions are prepared as follows (all at the same temperature):
- p_1 : Dissolve 12 g of glucose ($C_6H_{12}O_6$, M = 180) in 300 mL water
- p_2 : Dissolve 6 g of urea (CH_4N_2O, M = 60) in 300 mL water
- p_3 : Dissolve 17.1 g of sucrose ($C_{12}H_{22}O_{11}$, M = 342) in 300 mL water

Assuming all are non-electrolytes and solutions are dilute, the **correct** decreasing order of osmotic pressure is:
(1) $p_2 > p_1 > p_3$ (2) $p_1 > p_2 > p_3$
(3) $p_3 > p_1 > p_2$ (4) $p_2 > p_3 > p_1$

76. If an external opposite potential is applied in standard galvanic cell, then choose the **correct** statements when $E_{ext} > 1.1$ V
A. Electrons flow from Cu to Zn.
B. Current flows from Zn to Cu.
C. Zinc dissolves at the zinc electrode.
(1) A and B only (2) B and C only
(3) A and C only (4) A, B and C

77. Match **List-I** with **List-II**.

List-I (Conversion)		List-II (Number of Faraday required)	
(A)	1 mol of $K_2Cr_2O_7$ to Cr^{3+}	(I)	1 F
(B)	1 mol of I_2 to IO_3^-	(II)	6 F
(C)	1 mol of $FeSO_4$ to $Fe_2(SO_4)_3$	(III)	10 F
(D)	1 mol of NO to NO_3^-	(IV)	3 F

Choose the **correct** answer from the options given below:
(1) A-II; B-III; C-I; D-IV
(2) A-II; B-I; C-IV; D-III
(3) A-III; B-I; C-II; D-IV
(4) A-I; B-III; C-IV; D-II

78. An example of pseudo first order reaction is:
(1) Thermal decomposition of HI on gold surface
(2) Hydrogenation of ethene
(3) Inversion of cane sugar
(4) Decomposition of N_2O_5

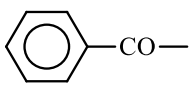
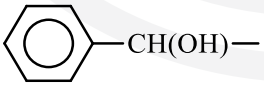
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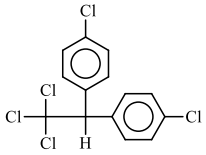
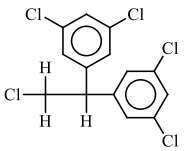
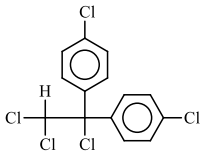
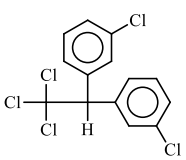
79. Among 3d series elements, enthalpy of atomization is maximum for:
(1) Mn (2) V
(3) Cr (4) Ti

80. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: MnO_4^- is typically more intensely coloured (purple) than MnO_4^{2-} (green).
Reason R: The oxidation state of Mn is higher in MnO_4^- than in MnO_4^{2-}.
In the light of above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A
(2) Both A and R are true but R is not the correct explanation of A
(3) A is true but R is false
(4) A is false but R is true

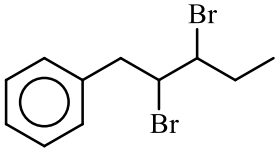
81. Most of the world wide production of phenol is from:
(1) Anisole (2) Benzophenone
(3) Cumene (4) Cresol

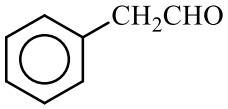
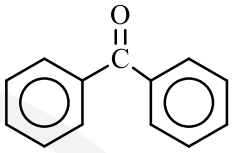
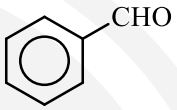
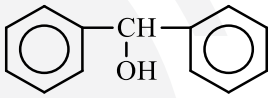
82. Which of the following compounds was once used as a general anaesthetic in surgery?
(1) $CHCl_3$ (2) CCl_4
(3) $COCl_2$ (4) CH_2Cl_2

83. Iodoform reaction with sodium hypoiodite is used for the detection of which group in organic compounds?
(1) $CH_3CH(OH)–$ (2) $CH_3CH_2CO–$
(3)  (4) 

84. The **correct** structure of DDT is:
(1)  (2) 
(3)  (4) 

85. Deficiency disease caused due to deficiency of vitamin B_2 is:
(1) Rickets (2) Xerophthalmia
(3) Cheilosis (4) Scurvy

86. IUPAC name of  is:
(1) 2,3-Dibromo-1-phenylpentane
(2) 3,4-Dibromo-5-phenylpentane
(3) 2,3-Dibromopentylbenzene
(4) 1-Phenyl-2,3-dibromopentane

87. Tautomerism is possible in:
(1) 
(2) 
(3) 
(4) 

88. Odd electron species among the following is:
(1) NO_2 (2) BeH_2
(3) SCl_2 (4) PCl_5

89. Match **List-I** with **List-II**.

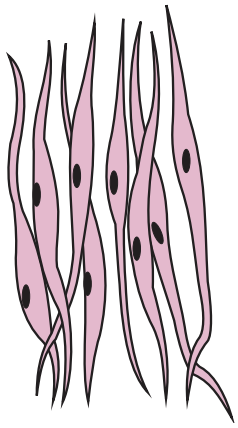
List-I		List-II	
(A)	$Xe^+PtF_6^-$	(I)	Colourless explosive solid.
(B)	XeF_2	(II)	Can be prepared by reacting XeF_4 and O_2F_2 at 143 K.
(C)	XeF_6	(III)	On hydrolysis gives Xe, HF and O_2
(D)	XeO_3	(IV)	First compound of xenon

Choose the **correct** answer from the options given below:
(1) A-IV; B-II: C-III; D-I
(2) A-IV; B-III; C-II; D-I
(3) A-I; B-III; C-II; D-IV
(4) A-I; B-II; C-III; D-IV

90. An electron has kinetic energy 150.5 eV. Its de Broglie wavelength (in Å) is closest to:
[Given: $\sqrt{150.5} = 12.27$]
(1) 1.00 Å (2) 0.50 Å
(3) 0.10 Å (4) 0.25 Å

[Contd...

91. Which of the following describes the diagram given below **accurately**?



(1) Cylindrical, striated, unbranched, and voluntary.
(2) Flat, irregularly shaped with loose intercellular matrix, and voluntary.
(3) Cylindrical, striated, branched with intercalated discs, and involuntary.
(4) Fusiform, non-striated and involuntary.

92. Root hairs are:
(1) unicellular elongations of collenchymatous cells.
(2) multicellular secretory structures.
(3) unicellular elongations of epidermal cells.
(4) found only in monocot plants.

93. Which of the following statements are **correct** regarding the structure of chloroplasts?
A. Chloroplasts have an outer membrane and an inner membrane.
B. The outer membrane forms flattened sacs called thylakoids.
C. Grana are made up of stacks of thylakoids.
D. The ribosomes of the chloroplasts are larger than the cytoplasmic ribosomes.

Choose the *most appropriate* answer from the options given below:
(1) A, B and C only (2) A and C only
(3) B and D only (4) C and D only

94. Identify the **incorrect** statement from the following.
(1) There is a definite region in *E. coli* DNA where the replication originates.
(2) A new strand always gets synthesised in 3' → 5' direction.
(3) DNA polymerases on their own cannot initiate the process of replication.
(4) Discontinuous strands are formed on template strand with polarity 5' → 3'.

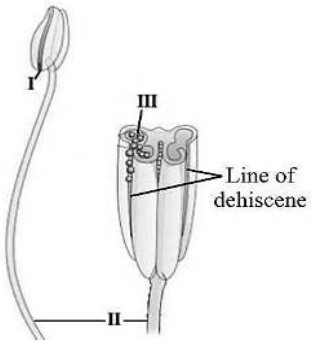
95. By the end of 12 weeks of pregnancy, which of the following occurs?
(1) Major organ systems are formed
(2) Foetus becomes fully developed
(3) Body is covered with fine hair
(4) Eyelids separate

96. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Leydig cells	(I)	Synthesize and secrete androgens
(B)	Sertoli cells	(II)	Formed after the first meiotic division of primary oocyte
(C)	First polar body	(III)	Provide nutrition to the male germ cells
(D)	Fimbriae	(IV)	Finger-like projections that help in collection of the ovum

Choose the **correct** answer from the options given below:
(1) A-I, B-III, C-II, D-IV
(2) A-IV, B-II, C-III, D-I
(3) A-IV, B-III, C-II, D-I
(4) A-I, B-III, C-IV, D-II

97. Identify the parts marked as **I, II,** and **III** in the diagram and choose the appropriate option.



(1) I-Anther, II-Filament, III-Endothecium
(2) I-Anther, II-Style, III-Pollen grains
(3) I-Anther, II-Style, III-Megasporangium
(4) I-Anther, II-Filament, III-Pollen grains

98. Which of the following statements regarding Prophase I of meiosis is **incorrect**?
(1) The synaptonemal complex forms in zygotene.
(2) Dissolution of the synaptonemal complex occurs during diplotene.
(3) During pachytene the four chromatids of each bivalent chromosome becomes distinct.
(4) In oocytes of some invertebrates, diplotene can last for months or years.

[Contd...

99. Which of the following is **true** about light reactions in photosynthesis?
(1) Light reactions occur in the stroma of chloroplasts.
(2) The LHC are made up of hundreds of pigment molecules bound to proteins.
(3) Photosystem I absorbs light energy at 680 nm.
(4) Chlorophyll *a* absorbs light and transfer the energy to chlorophyll *b*.

100. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Taxonomic studies are useful for agriculture, forestry, and industry.
Reason R: Taxonomy helps classify organisms based on evolutionary relationships.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

101. Identify the **correct** features of the human circulatory system:
A. The heart is mesodermally derived and situated in the thoracic cavity.
B. The SAN is located in the lower-left corner of the right atrium.
C. Action potential from AVN is conducted to the ventricular side by the AV bundle.
D. The tricuspid valve prevents the backflow of blood from the pulmonary artery into the right ventricle.
(1) A and B (2) A and C
(3) B and D (4) A and D

102. A patient is diagnosed with a malignant tumor. Which of the following properties is most likely exhibited by these tumor cells?
(1) High degree of contact inhibition.
(2) Confinement to their original location without causing surrounding tissue damage.
(3) Metastasis and competitive starvation of normal cells.
(4) Inability to undergo neoplastic transformation.

103. In the transport of oxygen, what percentage is carried by RBCs in the blood?
(1) 3 per cent (2) 20-25 per cent
(3) 70 per cent (4) 97 per cent

104. Consider the following statements.
A. rRNA provides the template for synthesis of proteins.
B. tRNA brings amino acids and reads the genetic code.
C. RNA polymerase binds to promoter and initiates transcription.
D. A segment of DNA coding for polypeptide is called intron.
Choose the *most appropriate* answer from the options given below:
(1) Only A and C are incorrect
(2) Only A and B are correct
(3) Only A, B and C are incorrect
(4) Only B and C are correct

105. Match **List-I** with **List-II**.

List-I (ECG peaks)		**List-II (Electrical activity)**	
(A)	P-wave	(I)	Ventricles return to normal state (Repolarisation)
(B)	QRS complex	(II)	End of ventricular systole
(C)	T-wave	(III)	Depolarisation of atria
(D)	End of T-wave	(IV)	Depolarisation of ventricles

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-II, D-I
(2) A-III, B-IV, C-I, D-II
(3) A-IV, B-III, C-II, D-I
(4) A-IV, B-III, C-I, D-II

106. In a sarcomere, what constitutes the 'A' band and 'I' band respectively?
(1) 'A' band contains thick myosin filaments; 'I' band contains thin actin filaments.
(2) 'A' band contains only actin filaments; 'I' band contains only myosin filaments.
(3) 'A' band contains a central elastic 'Z' line; 'I' band contains a fibrous 'M' line.
(4) 'A' band is the part not overlapped by thin filaments; 'I' band contains both actin and myosin.

107. Which of the following was a major reason for rapid population growth in India?
(1) Increase in maternal mortality rate.
(2) Increase in infant mortality rate.
(3) Decline in death rate, MMR and IMR.
(4) Decrease in number of people in reproductive age.

108. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The cerebellum has a very convoluted surface.
Reason R: The convoluted surface provides additional space for many more neurons.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

109. Which of the following carbohydrates forms a helical secondary structure that can hold I_2 molecules, giving a blue colour?
(1) Cellulose (2) Glycogen
(3) Inulin (4) Starch

110. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Steroids	(I)	Secondary messenger
(B)	Iodothyronines	(II)	Cortisol, Testosterone
(C)	IP_3	(III)	Pituitary hormones
(D)	Proteins/Peptides	(IV)	Thyroid hormones

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-IV, D-I
(2) A-III, B-II, C-IV, D-I
(3) A-II, B-IV, C-I, D-III
(4) A-IV, B-III, C-I, D-II

111. Given below are two statements:
Statement I: The descending limb of the loop of Henle is permeable to water but almost impermeable to electrolytes.
Statement II: The ascending limb of the loop of Henle allows transport of electrolytes actively or passively but is impermeable to water.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I and Statement II both are correct
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

112. Select the **incorrect** match between the organism and its characteristic:
(1) *Macropus* → Viviparous
(2) *Ornithorhynchus* → Oviparous
(3) *Pteropus* → Flying fox
(4) *Delphinus* → Cold-blooded

113. Which primary lymphoid organ reduces to a very small size by the time puberty is attained?
(1) Bone marrow (2) Spleen
(3) Thymus (4) Lymph node

114. Select the **incorrect** statement regarding the ribs in a human body:
(1) There are 12 pairs of ribs, each having two articulation surfaces on its dorsal end.
(2) The first seven pairs of ribs are called true ribs as they are ventrally connected to the sternum with the help of hyaline cartilage.
(3) The 8th, 9th, and 10th pairs of ribs do not articulate directly with the sternum and are called floating ribs.
(4) Thoracic vertebrae, ribs, and sternum together form the rib cage.

115. Find the **correct** statement from the options given below.
(1) Once the BOD of sewage or waste water is increased significantly, the flocs sediment to form activated sludge.
(2) A small part of the activated sludge is pumped back for primary treatment where it serves as the inoculum.
(3) In anaerobic sludge digesters the bacteria and the fungi in the sludge get digested.
(4) In anaerobic sludge digesters fungi grow anaerobically.

116. Which of the following are **true** about algae?
A. *Chlorella* is used as a food supplement for space travellers.
B. Pyrenoids in algae contain protein besides starch.
C. Certain red algae provide hydrocolloids such as carrageen.
D. Green algae usually have a cell wall made of an outer layer of cellulose and an inner layer of pectose.
Choose the *most appropriate* answer from the options given below:
(1) A, B and C only (2) A, B and D only
(3) A and C only (4) A, C and D only

117. Match **List-I** with **List-II**:

List-I		List-II	
(A)	Habitat loss and fragmentation	(I)	Steller's sea cow
(B)	Over-exploitation	(II)	Extinction of plant leading to the extinction of its pollinator
(C)	Alien species invasions	(III)	Cutting of Amazon rain forest for cultivation of soya beans
(D)	Co-extinctions	(IV)	The Nile perch introduced into Lake Victoria in east Africa

Choose the **correct** answer from the options given below:
(1) A-IV, B-I, C-II, D-III
(2) A-IV, B-II, C-I, D-III
(3) A-III, B-I, C-IV, D-II
(4) A-IV, B-III, C-II, D-I

118. In a Polymerase Chain Reaction (PCR), which organism is the source of the thermostable enzyme?
(1) *Escherichia coli*
(2) *Salmonella typhimurium*
(3) *Thermus aquaticus*
(4) *Agrobacterium tumifaciens*

119. Which statements about polysomes are **correct**?
A. Found in eukaryotes only
B. Contain mRNA and ribosomes
C. Contain pigments
D. Play a role in protein synthesis
Choose the *most appropriate* answer from the options given below:
(1) A only (2) B and D only
(3) A, B, and C only (4) A, B, C, and D

120. Read the following statements and find out the **incorrect** statements.
A. Majority of flowering plants use a range of animals as pollinating agents.
B. Ants and moths are common pollinating agents.
C. Birds are the dominant biotic pollination agents.
D. Even reptiles have been reported as pollinators in some species.
E. Often flowers of animal pollinated plants are specifically adapted for a particular species of animal.
Choose the *most appropriate* answer from the options given below.
(1) C only (2) A and B only
(3) B only (4) A, B and C only

121. Given below are two statements:
Statement I: All vertebrates are chordates but all chordates are not vertebrates.
Statement II: The notochord is replaced by a cartilaginous or bony vertebral column in the adult echinoderms.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I and Statement II both are correct
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

122. Match **List-I** with **List-II:**

List-I		List-II	
(A)	Alfred Sturtevant	(I)	Experimental verification of chromosomal theory of inheritance
(B)	Henking	(II)	X-body
(C)	Walter Sutton and Theodore Boveri	(III)	Chromosomal theory of inheritance
(D)	Thomas Hunt Morgan	(IV)	Mapped position of genes

Choose the **correct** answer from the options given below:
(1) A-I, B-III, C-IV, D-II
(2) A-IV, B-II, C-III, D-I
(3) A-IV, B-I, C-III, D-II
(4) A-III, B-II, C-IV, D-I

123. Sacred groves are found in:
(1) Khasi and Jaintia Hills in Maharashtra.
(2) Aravalli Hills of Madhya Pradesh.
(3) Western ghat regions of Karnataka.
(4) Sarguja, Chanda and Bastar areas of Meghalaya.

124. The age of the universe is approximately:
(1) 4.6 billion years
(2) 13.8 billion years
(3) 1 billion years
(4) 100 million years

125. The gizzard in cockroaches mainly helps in:
(1) Grinding of food particles
(2) Absorption of nutrients
(3) Digestion of proteins
(4) Storage of food

126. Given below are two statements:
Statement I: Hormones which interact with membrane-bound receptors normally enter the target cell and generate second messengers.
Statement II: Steroid hormones and iodothyronines interact with intracellular receptors and regulate gene expression.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

127. Where is ATP synthase located in the mitochondria?
(1) Mitochondrial matrix
(2) Outer mitochondrial membrane
(3) Inner mitochondrial membrane
(4) Cytoplasm

128. Select the **correct** statements regarding the process of photorespiration in C_3 plants.
(1) Involves the fixation of only CO_2 by RuBisCO.
(2) Reduces the efficiency of photosynthesis.
(3) Results in the production of ATP.
(4) Occurs when O_2 binds to PGA.

129. Identify from the following microorganisms which help in nitrogen fixation.
A. *Oscillatoria*
B. *Azotobacter*
C. *Anabaena*
D. *Glomus*
E. *Azospirillum*
Choose the *most appropriate* answer from the options given below:
(1) A, B, C and E only
(2) A and C only
(3) A, B and E only
(4) D only

130. Fatty acids enter the respiratory pathway:
(1) as glucose.
(2) after being converted into glycerol.
(3) after being converted to acetyl CoA.
(4) after being converted to pyruvate.

131. In the study of protein structure, which of the following refers to the architecture of a protein involving the spatial arrangement of more than one polypeptide or subunit?
(1) Primary structure
(2) Secondary structure
(3) Tertiary structure
(4) Quaternary structure

132. MTPs are considered relatively safe up to which period of pregnancy?
(1) Upto the first 6 weeks only
(2) Upto the first trimester
(3) Upto the second trimester
(4) Upto 30 weeks

133. Which specific muscles help in increasing the strength of inspiration and expiration in humans?
(1) Primary bronchi muscles
(2) Cardiac muscles
(3) Additional muscles in the abdomen
(4) Only the diaphragm

134. How many amino acids will be coded by the mRNA sequence?
5' AUGUUUGUGUUUUUCUAAA 3'
(1) Three only (2) Five
(3) Four only (4) Two only

135. Which of the following is promoted by ethylene?
(1) Leaf formation
(2) Root hair formation
(3) Lateral shoot growth
(4) Chloroplast in leaves

136. Which of the following molecular diagnostic techniques relies on the principle of antigen-antibody interaction?
(1) Polymerase Chain Reaction
(2) Autoradiography
(3) Recombinant DNA technology
(4) Enzyme Linked Immuno-sorbent Assay

137. Select the **correct** statements about viruses:
A. Viruses can only replicate inside the host cell.
B. Bacteriophages are viruses that infect plant cells.
C. No virus contains both RNA and DNA.
D. Viruses lack a inert crystalline structure outside the host cell.
Choose the *most appropriate* answer from the options given below:
(1) A and C only (2) A, B, and C only
(3) A, C, and D only (4) A, B, C, and D

138. Identify the **incorrect** statement about dicotyledonous roots.
(1) The epiblema produces root hairs.
(2) The cortex consists of thick-walled cells.
(3) The endodermis has casparian strips.
(4) The pericycle initiates lateral roots.

139. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Chromatin	(I)	Nuclear envelope
(B)	Satellite	(II)	Non-staining secondary constrictions
(C)	Nucleolus	(III)	Site of active ribosomal RNA synthesis
(D)	Perinuclear space	(IV)	Stained by the basic dyes

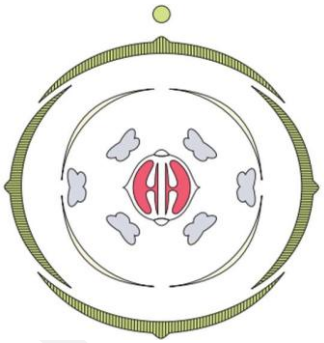
Choose the **correct** answer from the options given below:
(1) A-IV, B-II, C-III, D-I
(2) A-I, B-II, C-IV, D-III
(3) A-IV, B-I, C-II, D-III
(4) A-II, B-IV, C-III, D-I

140. Which of the following is a characteristic feature of the class Aves?
(1) Heart is usually three-chambered but four-chambered in some.
(2) Skin is dry with abundant exocrine glands.
(3) The digestive tract has additional chambers, the crop and gizzard.
(4) They are poikilothermous animals.

141. Which of the following **correctly** differentiates between monocot roots and dicot stems?
(1) Monocot roots have a well-developed pith, whereas dicot stems do not have a pith.
(2) Monocot roots do not undergo secondary growth, while dicot stems typically develop a vascular cambium for secondary growth.
(3) Dicot stems have a polyarch arrangement of vascular bundles.
(4) Monocot roots have an epidermis covered with a thick cuticle.

142. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The thickness of the walls of ventricles is much greater than that of the atria.
Reason R: Size of ventricles is greater than atrium.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

143. Select the **correct** floral formula representation of the given part for the floral diagram given below.



(1) A = (2 + 2)
(2) C = (2 + 2)
(3) G = 2 + 2
(4) K = 2+2

144. Identify the **incorrect** statement from the following.
(1) Regulation of transcription is the primary step for gene expression.
(2) The tertiary structure of tRNA looks like a clover-leaf.
(3) tRNAs are specific for each amino acid.
(4) The anticodon loop of tRNA contains three bases that pair complementary with the codon on the mRNA template.

145. Match **List-I** with **List-II.**

List-I		List-II	
(A)	*Volvox*	(I)	Example of isogamous reproduction with flagellated gametes
(B)	*Ulothrix*	(II)	Example of anisogamous reproduction
(C)	*Spirogyra*	(III)	Example of oogamous reproduction
(D)	*Eudorina*	(IV)	Example of isogamous reproduction with non-flagellated gametes

Choose the **correct** answer from the options given below:
(1) A-III, B-I, C-IV, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-II, B-III, C-IV, D-I
(4) A-II, B-I, C-III, D-IV

146. Transgenic cow Rosie, produced in 1997, was significant because her milk contained:
(1) Alpha-1-antitrypsin to treat emphysema.
(2) High levels of insulin for diabetic patients.
(3) Human alpha-lactalbumin, making it nutritionally more balanced for human babies.
(4) A specific vaccine for the polio virus.

147. Which of the following sets of structures represents convergent evolution?
(1) Forelimbs of whales, bats, cheetah, and human.
(2) Thorns of *Bougainvillea* and tendrils of *Cucurbita.*
(3) Eye of the octopus and the eye of mammals.
(4) Vertebrate hearts and vertebrate brains.

148. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Trophoblast	(I)	Contains certain cells called stem cells which have the potency to give rise to all tissues
(B)	Inner cell mass	(II)	A new membrane formed by the secondary oocyte surrounding it
(C)	Antrum	(III)	Outer layer of blastocyst that gets attached to the endometrium
(D)	Zona pellucida	(IV)	A fluid-filled cavity that characterizes a tertiary follicle

Choose the **correct** answer from the options given below:
(1) A-III, B-I, C-IV, D-II
(2) A-I, B-IV, C-II, D-III
(3) A-III, B-I, C-II, D-IV
(4) A-II, B-III, C-I, D-IV

149. According to David Tilman's experiment:
(1) plots with more species showed more year-to-year variation in total biomass.
(2) key species drive major ecosystem functions.
(3) rich biodiversity is not essential for ecosystem health.
(4) increased diversity contributed to higher productivity.

150. Identify the **correct** statements regarding respiratory volumes:
A. Tidal Volume is approximately 500 mL in a healthy man.
B. Residual Volume averages 1200 mL to 1500 mL.
C. Inspiratory Reserve Volume averages 2500 mL to 3000 mL.
D. Vital Capacity is the sum of RV, ERV, TV, and IRV.
(1) A and C
(2) A, B, and C
(3) A, C, and D
(4) B and D

151. Identify the **incorrect** statement regarding biological control agents.
(1) *Bacillus thuringiensis* produces toxins harmful to butterfly caterpillars.
(2) *Trichoderma* species are effective in controlling plant pathogens.
(3) Dragonflies help control the small fishes population.
(4) Ladybirds help control the aphid populations.

152. Inflammation of the glomeruli of the kidney is a condition known as:
(1) Renal calculi
(2) Uremia
(3) Glomerulonephritis
(4) Glycosuria

153. Choose the **incorrectly** matched pair from the following.
(1) Compositae - Sunflower
(2) Malvaceae - Cotton
(3) Gramineae - *Triticum*
(4) Brassicaceae - Lady's finger

154. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The M phase is the shortest phase of the cell cycle.
Reason R: During the M phase, cell division and chromosome replications occurs.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

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155. Read the following statements and identify the **correct** statement.
(1) Linnaeus' system of classification was based on gynoecium structure.
(2) Artificial classification includes criteria like internal features.
(3) Natural classification includes criteria such as embryology and phytochemistry.
(4) Numerical taxonomy does not use observable characteristics.

156. Parasympathetic neural signals affect the cardiac output by:
(1) increasing the rate of heart beat.
(2) increasing the strength of ventricular contraction.
(3) decreasing the rate of heart beat and speed of conduction.
(4) increasing the blood pressure.

157. During the resting state of a neuron, how does the axoplasm's ionic composition contrast with the fluid outside the axon?
(1) High concentration of Na^+ inside and high concentration of K^+ outside.
(2) Low concentration of K^+ and negatively charged proteins inside.
(3) High concentration of K^+ and negatively charged proteins inside, and low concentration of Na^+ inside.
(4) Equal concentrations of Na^+ and K^+ across the membrane maintained by passive diffusion.

158. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: For the multiplication of any alien piece of DNA in an organism, it needs to be linked to a specific sequence known as rop.
Reason R: The origin of replication is a sequence responsible for initiating replication and controlling the copy number of the linked DNA.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

159. During the commercial production of genetically engineered human insulin by Eli Lilly in 1983, what major challenge had to be overcome?
(1) Extracting insulin from slaughtered cattle without causing allergies.
(2) Getting the recombinant insulin assembled into a mature form by creating disulfide bonds between chain A and chain B.
(3) Removing the extra C peptide from the mature insulin chain within the *E. coli* cells.
(4) Fusing the protoplasts of tomato and potato to create a bioreactor plant.

160. Identify the **correct** statements regarding the Adrenal Cortex:
A. The outer layer is called zona glomerulosa.
B. Cortisol is the main mineralocorticoid in our body.
C. Aldosterone acts mainly at the renal tubules for Na^+ and water reabsorption.
D. Glucocorticoids inhibit gluconeogenesis and stimulate cellular uptake of amino acids.
(1) A and C (2) A, B, and C
(3) B and D (4) C and D

161. Select the **mismatched** pair from the following.
(1) Female heterogamety – Birds.
(2) Male heterogamety – *Drosophila*
(3) Female heterogamety – Humans
(4) Male heterogamety – Grasshopper

162. The special venous connection between the liver and intestine in frogs is called:
(1) Renal portal system
(2) Coronary circulation
(3) Pulmonary portal system
(4) Hepatic portal system

163. Which of the following respiratory capacities includes the Residual Volume?
(1) Vital Capacity
(2) Inspiratory Capacity
(3) Functional Residual Capacity
(4) Expiratory Capacity

164. How many of the following are **not** examples of terrestrial ecosystems?
Forest, pond, aquarium, grassland, crop fields, estuary, lake.
Choose the *most appropriate* answer from the options given below:
(1) Seven (2) Four only
(3) Five only (4) Three only

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165. Given below are two statements:

Statement I: Majority of the parasites harm the host.

Statement II: Many marine fish are infested with ectoparasitic copepods.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

166. According to the theory of special creation, the age of the earth is about:

(1) 4 million years
(2) 4000 years
(3) 4 billion years
(4) 40,000 years

167. The drug commonly called smack is chemically formed by which of the following processes?

(1) Methylation of cannabinoids extracted from *Cannabis sativa.*
(2) Acetylation of morphine extracted from the latex of *Papaver somniferum*.
(3) Glycosylation of cocaine extracted from *Erythroxylum coca*.
(4) Hydroxylation of alkaloids from *Atropa belladona*.

168. Which of the following is **incorrect** regarding the tricarboxylic acid (TCA) cycle?

(1) The TCA cycle occurs in the mitochondrial matrix.
(2) At one step ATP is produced directly by substrate-level phosphorylation.
(3) $FADH_2$ and NADH are produced in the TCA cycle.
(4) Oxygen is used in the TCA cycle.

169. Restriction endonuclease *Hind* II always cuts DNA molecules at a particular point by recognizing a specific sequence of:

(1) Four base pairs
(2) Six base pairs
(3) Eight base pairs
(4) Ten base pairs

170. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: In cortical nephrons, the *vasa recta* is absent or highly reduced.

Reason R: The loop of Henle in cortical nephrons is very short and extends only very little into the medulla.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

171. Read the following and select how many statements are **correct**.

a. More than 15,500 species world-wide are facing the threat of extinction.
b. The colonisation of tropical Pacific Islands by humans is said to have led to the extinction of more than 2,000 species of native birds.
c. Among plants, the number of algal species is more than the angiosperms species.
d. In general, loss of biodiversity in a region may lead to a decline in plant production.

Choose the *most appropriate* answer from the options given below:

(1) One only
(2) Two only
(3) Three only
(4) Four

172. How many of the following are examples of synovial joints?

A. Sutures in the flat skull bones.
B. Joint between the adjacent vertebrae.
C. Pivot joint between atlas and axis.
D. Saddle joint between carpal and metacarpal of the thumb.
E. Ball and socket joint between humerus and pectoral girdle.

(1) Two
(2) Three
(3) Four
(4) Five

[Contd...

173. Find the **incorrect** statement(s) among the following.
A. Earthworm breaks down detritus into smaller particles.
B. Identification and enumeration of plant and animal species of an ecosystem gives its species composition.
C. The rate of energy consumption is called productivity.
D. Humification leads to accumulation of a light coloured crystal substance called humus.
Choose the *most appropriate* answer from the options given below.
(1) A and C only (2) A, C and D only
(3) C and D only (4) All A, B, C and D

174. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The sewage water is treated to reduce BOD before being released into natural water bodies.
Reason R: The greater the BOD of waste water, more is its polluting potential.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

175. The process in which a plant's cells lose the ability to divide and specialize into specific functions is called:
(1) growth. (2) dedifferentiation.
(3) differentiation. (4) redifferentiation.

176. Identify the **correct** statements from the following.
A. At present, the world is facing the sixth mass extinction of species.
B. There is more solar energy available in the temperate which contributes to higher productivity and greater diversity.
C. Species area relationship was explained by Alexander von Humboldt.
D. The Eastern Ghats have a greater amphibian species diversity than the Western Ghats.
Choose the correct answer from the options given below:
(1) A and B (2) B and C
(3) A and C (4) B and D

177. A child's blood group is 'O'. The genotype of mother and father blood groups can be:
(1) B (heterozygous) and B (homozygous) respectively.
(2) A (heterozygous) and B (homozygous) respectively.
(3) B (homozygous) and O (homozygous) respectively.
(4) A (heterozygous) and A (heterozygous) respectively.

178. Which of the following statements is **correct** for biodiversity of India?
(1) India, with much of its land area in the tropical latitudes, has more than 1,500 species of birds.
(2) Its share of the global species diversity is an impressive 18.1 per cent.
(3) Nearly 45,000 species of plants and twice as many of animals have been recorded from India.
(4) India has only 4.2 per cent of the world's land area.

179. Which of the following statements are **correct** about the leafy stage in mosses?
(I) It develops from the secondary protonema.
(II) It consists of upright, slender axes bearing spirally arranged leaves.
(III) It consists of a creeping structure.
(IV) It bears the sex organs of the moss.
Choose the *most appropriate* answer from the options given below:
(1) I, II and III only
(2) I, II and IV only
(3) I and III only
(4) I, III and IV only

180. Which of the following **correctly** describes the components of a nucleotide?
(1) A heterocyclic compound and a monosaccharide only.
(2) A substituted purine/pyrimidine, a hexose sugar, and a phosphoric acid.
(3) A heterocyclic compound, a monosaccharide (pentose), and a phosphoric acid.
(4) A nitrogenous base attached directly to a phosphoric acid.

■■■

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